

EduGenie Technology Stack

Technology Overview

Layer	Technology	Purpose
Programming Language	Python	Backend development
Backend Framework	FastAPI	REST API development
AI Platform	Google Gemini	Educational content generation
AI SDK	google-genai	Connecting Python with Gemini
Frontend Structure	HTML	Web page structure
Frontend Styling	CSS	Layout and responsive design
Frontend Logic	JavaScript	Calling backend APIs
Validation	Pydantic	Request validation
Configuration	python-dotenv	Loading environment variables
Testing	Pytest	Automated testing
Version Control	Git	Source code management
Code Hosting	GitHub	Project hosting
Documentation	Markdown and PDF	Project reports
Optional Deployment	Docker	Application packaging

Backend Files

```
app/
├── __init__.py
├── main.py
├── schemas.py
└── ai_service.py
```